

3(a)(i)	any three from: 1 glucose diffuses into the cell (through, A / transport protein) ; 2 <i>idea that</i> the enzyme / B, loses phosphate group ; 3 (dephosphorylated) B binds to lactose permease ; 4 causes, shape change / allosteric change, to lactose permease ; 5 lactose permease cannot function (so lactose cannot enter) ;	3
3(a)(ii)	any two from: 1 no / less, (protein) synthesis of, lactose permease / β-galactosidase / transacetylase or no / less, transcription / expression, of (named structural) genes (of lac operon) ; 2 no / less, ATP / energy, wasted ; 3 no / fewer, amino acids wasted ; 4 glucose is respired more easily / AW or fewer steps / less energy, required to break down glucose or lactose has to be broken down (to glucose and galactose) ;	2

Question	Answer	Marks
3(a)(iii)	any three from: 1 splits into / forms, two, triose phosphate / TP (molecules) ; 2 (TP) oxidised / dehydrogenated, to pyruvate ; 3 ATP production ; 4 substrate-linked phosphorylation ; 5 production of reduced NAD ;	3

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3(b)(i)	any two from: <i>lactose diet</i> 1 increases (over six days) ; <i>no lactose diet</i> 2 small increase / increases slightly / little change / stays around 50% (over six days) ; 3 comparative data quote – 2 percentages of either diet over time or percentage of each diet on one day ; <table border="1"> <thead> <tr> <th>time / days</th><th colspan="2">percentage lac⁺ ± 0.5 mm</th></tr> <tr> <th></th><th>lactose diet</th><th>no lactose diet</th></tr> </thead> <tbody> <tr> <td>0</td><td>50</td><td>50</td></tr> <tr> <td>1</td><td>65.5</td><td>54</td></tr> <tr> <td>2</td><td>75.5</td><td>56.5</td></tr> <tr> <td>3</td><td>83.5</td><td>60</td></tr> <tr> <td>4</td><td>84</td><td>59</td></tr> <tr> <td>5</td><td>85</td><td>57.5</td></tr> <tr> <td>6</td><td>86</td><td>57</td></tr> </tbody> </table>	time / days	percentage lac ⁺ ± 0.5 mm			lactose diet	no lactose diet	0	50	50	1	65.5	54	2	75.5	56.5	3	83.5	60	4	84	59	5	85	57.5	6	86	57	2
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3(b)(ii)	any two from: 1 lactose is the selection pressure ; 2 (<i>lac</i>⁺), can metabolise / take up / AW, lactose (so survive) or <i>lac</i>⁻, cannot metabolise / take up / AW, lactose (so die) ; 3 AVP ; e.g. <i>ref.</i> glucose has run out	2

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